

CHM 320: ADVANCED INORGANIC CHEMISTRY TEXTBOOK



CHM 320 Advanced Inorganic Chemistry
textbook

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CHAPTER OVERVIEW

1: Introduction to Inorganic Chemistry

This chapter introduces some history and context about the field of Inorganic Chemistry.

[1.1: Inorganic vs Organic Chemistry](#)

[1.2: History of Inorganic Chemistry](#)

[1.3: Perspectives](#)

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1.1: Inorganic vs Organic Chemistry

The division between the fields of Inorganic and Organic chemistry has become blurred. For example, let's look at one of the major classes of catalysts used for organic synthesis reactions: organometallic catalysts (Figure 1.1.1). Organometallic catalysts like these, and all organometallic compounds, contain metals that are bonded to carbon or carbon-containing molecules. So, are they "inorganic" because they contain metals, or "organic" because they contain carbon? These illustrate that clear divisions between organic and inorganic chemistry do not exist. Further, metal ions are common in biology and so the idea that metals are "inorganic" and thus classed as "non-living or non-biological" is incorrect. A canonical example is the organometallic catalyst adenosylcobalamin, which is an important biological cofactor containing a cobalt (Co) ion (Figure 1.1.1, right) and a cobalt-carbon bond.

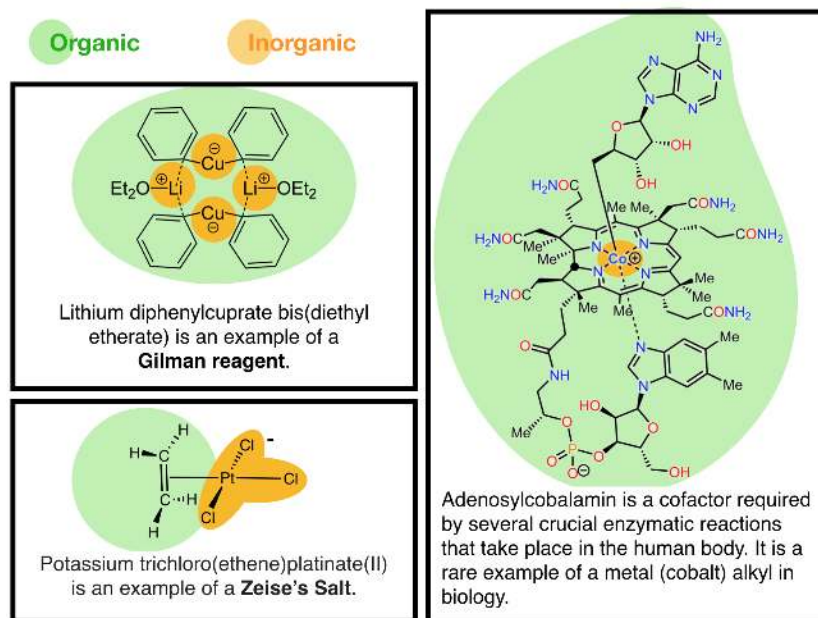


Figure 1.1.1: Some examples of organometallic catalysts. These compounds catalyze organic reactions or biochemical reactions and they are compounds that contain both carbon and metals. These compounds are examples of molecules that cannot be defined only as organic molecules or only as inorganic molecules. Adenosylcobalamin is an example of an organometallic catalyst that is present in biology, further illustrating that "inorganic" metals are important cofactors in biology. This image is based on information on the Wikipedia article on Organometallic Chemistry and is created from images found there; attribution to images created by [Alsosaid1987](#), [AdoCbl-ColorCoded](#), [CC BY-SA 4.0](#) and [Smokefoot](#), [Zeise'sSalt](#), [CC BY-SA 3.0](#). (Kathryn Haas; CC-BY-SA)

Some of the subfields of Inorganic Chemistry focus on electrical conductivity of inorganic materials (i.e., conduction, superconduction, and semiconduction) and on the study of optical and electronic properties of inorganic nanomaterials. Electrical conductivity is a canonical property of metals, but carbon-based materials also demonstrate electrical conductivity. For example, carbon nanotubes conduct electricity through their extended conjugated π systems. Fullerenes, of which the most famous is Buckminsterfullerene, or Buckeyball (C_{60}), demonstrate interesting properties that are similar to nanoparticles, and when combined with metals and crystallized can demonstrate superconductivity.

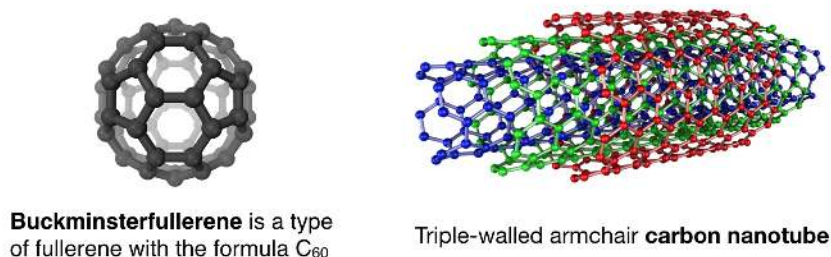


Figure 1.1.2: This figure is created from information found on the Wikipedia articles for [Buckminsterfullerene](#) and [carbon nanotubes](#). Attribution [Eric Wieser](#), [Multi-walled Carbon Nanotube](#), [CC BY-SA 3.0](#). (Kathryn Haas; CC-BY-SA)

Although carbon nanotubes and fullerenes are allotropes of carbon, their material properties are somewhat foreign to many organic chemists, who traditionally have focused on smaller organic molecules having very different properties. However, these properties are familiar to inorganic chemists. Thus, inorganic chemists have embraced these molecules as "inorganic" due to the fact that they behave more like inorganic materials than smaller organic molecules. This class of carbon-based molecules serves as another example of molecules that are not perfectly matched to the traditional definitions of "organic" and "inorganic" chemistry. Certainly, the future will hold more and more examples of molecules that do not fit into the traditional disciplines of chemistry.

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